Counterfactual Temporal Reasoning with Event Intervals

Anonymous ACL submission

Abstract

Some mathematical, multi-hop, commonsense, and narrative reasoning tasks require models to track event order, state changes, and temporal dependencies. Existing evaluations usually focus on final answer accuracy, but a correct answer does not necessarily imply that the model has represented event order and temporal relations. Prompting based methods may exploit surface temporal cues while leaving the underlying event structure implicit.

We study this issue through counterfactual event order shifts, which preserve most non-temporal content but change the relation between target events. A temporally reliable model should update its relation prediction, and should update the final answer when the answer depends on the changed relation. However, prompting based baselines can remain insensitive to such shifts because their reasoning is expressed only as unconstrained text.

To address this, we propose Kairos, an interval based framework for counterfactual temporal reasoning. Kairos maps event mentions into latent start and end coordinates, decodes pairwise temporal relations through geometry guided features, and uses the resulting event relation graph to rerank candidate answers. We further introduce counterfactual temporal supervision to train relation predictions to update under order changing perturbations. Experiments suggest that explicit event interval representation improves robustness over prompting based baselines, especially under counterfactual shifts.

1 Introduction

Large language models have become a general reasoning interface for natural language tasks (Brown et al., 2020; Ouyang et al., 2022; Achiam et al., 2023; Touvron et al., 2023; Yang et al., 2024). Some reasoning tasks require models to understand time. In mathematical reasoning, a model may need to track state changes that unfold across steps. In multi-hop question answering, it may need to compare events across passages. In commonsense reasoning, it may need to infer which event must occur before another (Cobbe et al., 2021; Geva et al., 2021; Trivedi et al., 2022; Ho et al., 2020). These cases require more than fluent generation. They require a representation of event order, duration, overlap, and persistence.

Prompting based methods provide a convenient way to elicit reasoning from large language models. Direct prompting asks the model to answer from the input. Chain of thought prompting encourages intermediate rationales (Wei et al., 2022; Kojima et al., 2022). Self consistency aggregates multiple sampled rationales (Wang et al., 2023). These methods can improve answer accuracy, but their reasoning remains in free form language space. The rationale is not required to maintain event boundaries, compare temporal intervals, or preserve a consistent event relation graph.

This creates a gap between answer correctness and temporal understanding. A model may answer a standard temporal question correctly by relying on surface expressions such as before, after, during, and then. It may also exploit common event priors learned during pretraining. Such behavior can be sufficient for the original example, but it does not show that the model has represented the event structure needed for robust temporal reasoning.

We expose this gap with counterfactual event order shifts. These shifts change a target temporal relation while preserving most non-temporal content. For example, if an original query states that event e_i happens before event e_j, a counterfactual query may reverse this order. A temporally reliable model should update the relation prediction, and should update the final answer when the answer depends on this relation. However, prompting based models may preserve the original prediction because the perturbed query remains lexically similar. This reveals temporal structure inconsistency, where standard accuracy hides unstable event relation reasoning.

To address this issue, we propose Kairos, an interval based framework for counterfactual temporal reasoning. Kairos constructs conservative event and relation labels, maps event mentions into latent intervals, decodes an event relation graph, and reranks candidate answers with counterfactual temporal supervision that trains relation updates under order shifts.

The main contributions of our work are summarized as follows.

- We formalize temporal structure inconsistency as a failure mode where models answer standard temporal examples correctly but fail to update predictions under counterfactual event order shifts.
- We propose Kairos, an interval based temporal reasoning framework that maps event mentions into latent intervals, decodes an event relation graph, and uses this graph to guide answer selection.
- We introduce counterfactual temporal supervision, which treats event order perturbations as structural changes and trains relation predictions to update rather than remain invariant.
- Experiments across temporal reasoning settings show that Kairos improves robustness over prompting based baselines, while ablations identify the role of relation supervision, interval projection, counterfactual supervision, and graph based answer scoring.

2 Related Work

2.1 Temporal Reasoning in Language Models

Temporal reasoning builds on annotation schemes, Allen interval algebra, and neural relation extraction (Allen, 1983; Pustejovsky et al., 2003; UzZaman et al., 2013; Ning et al., 2019; Han et al., 2021; Ning et al., 2020; Chen et al., 2021). Large language models usually rely on prompting based reasoning (Wei et al., 2022; Wang et al., 2023), but rationales are not required to satisfy temporal consistency (Chu et al., 2024; Qiu et al., 2024). We add an explicit event relation graph on top of language model representations.

2.2 Counterfactual Robustness

Counterfactual evaluation tests whether models rely on robust patterns or superficial correlations (McCoy et al., 2019; Kaushik et al., 2020; Gardner et al., 2020; Ribeiro et al., 2020). We emphasize whether intermediate event relations update after order changing perturbations, not only whether final answers change.

2.3 Structured and Compositional Reasoning

Structured reasoning work shows that final answer accuracy may hide shallow shortcuts (Min et al., 2019; Sinha et al., 2019; Keysers et al., 2020; Yang et al., 2018; Trivedi et al., 2022). Kairos learns latent event intervals from language model representations and trains relation updates under counterfactual order shifts (Zhou et al., 2022; Xiong et al., 2024).

3 Temporal Structure Inconsistency

We use temporal structure inconsistency for cases where a model answers the original query correctly but fails to update when event order changes. This motivates Kairos to represent event boundaries, predict pairwise relations, and train those relations to update under order changing perturbations.

4 Kairos

Building on the above analysis, Kairos makes temporal structure explicit and trainable. It constructs event mentions and relation labels, projects events into latent intervals, decodes an event relation graph, and reranks candidate answers with counterfactual temporal supervision. A backbone language model proposes candidates; Kairos selects the answer most consistent with the predicted graph.

4.1 Event and Relation Construction

Given a query q, we construct event mentions and reliable relation labels. Verbal predicates and nominal event expressions form candidates; non-temporal candidates are removed to obtain $E = \{e_1, e_2, \ldots, e_n\}$. Relation labels come from explicit markers such as before, after, during, while, until, when, then, earlier, later, and followed by. Event pairs without reliable markers are excluded from the relation loss but may still contribute to answer supervision. Table 2 reports marker and relation-label coverage.

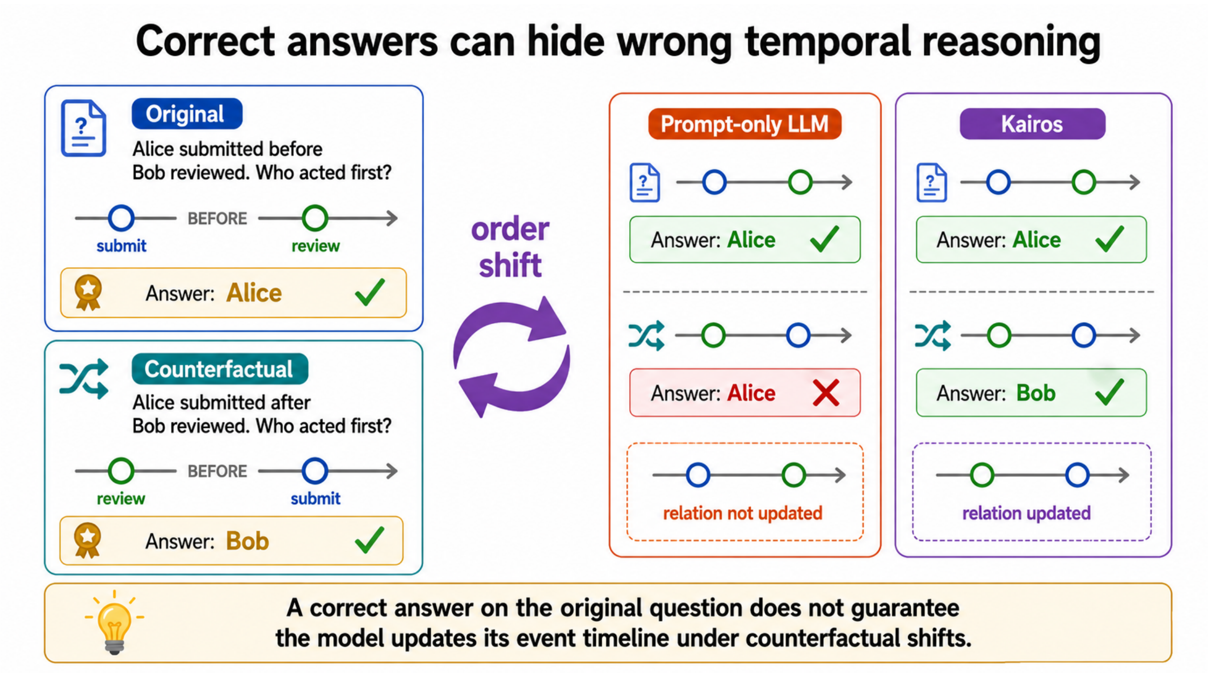

Figure 1: **Motivation.** A model can answer the original temporal question correctly while failing to update the underlying event relation after a counterfactual order shift. Kairos makes the relation update explicit and uses it to change answer selection when needed.

4.2 Event Interval Estimation

Given the event set $E = \{e_1, e_2, \ldots, e_n\}$, we encode each event mention with span pooled hidden states:

$$\mathbf{h}_i = \mathrm{Pool}_{t \in \mathrm{span}(e_i)} \mathbf{H}_t. \tag{1}$$

Each event is projected into a start coordinate and positive duration:

$$\begin{aligned} \hat{s}_i &= \mathbf{w}_s^\top \mathbf{h}_i, \\ \hat{d}_i &= \mathrm{softplus}(\mathbf{w}_d^\top \mathbf{h}_i), \quad \hat{e}_i = \hat{s}_i + \hat{d}_i. \end{aligned} \tag{2}$$

The interval is latent and represents relative temporal position within the query rather than absolute timestamps.

4.3 Geometry Guided Relation Decoding

Kairos decodes temporal relations by comparing interval endpoints. The precedence feature is

$$z_{\mathrm{precedes}}(i,j) = \hat{s}_j - \hat{e}_i, \tag{3}$$

and overlap is

$$z_{\mathrm{overlaps}}(i,j) = \min(\hat{s}_j - \hat{s}_i,\ \hat{e}_i - \hat{s}_j,\ \hat{e}_j - \hat{e}_i). \tag{4}$$

Interval features feed a relation classifier:

$$\mathbf{g}_{ij} = [\hat{s}_i, \hat{e}_i, \hat{s}_j, \hat{e}_j, \hat{s}_j - \hat{e}_i, \hat{s}_i - \hat{e}_j, \hat{d}_i, \hat{d}_j], \tag{5}$$

$$P_\theta(r_{ij} \mid q) = \mathrm{softmax}(W_r \mathbf{g}_{ij}). \tag{6}$$

Given relation labels $\mathcal{R}_q$,

$$\mathcal{L}_{\mathrm{rel}} = - \sum_{(i,j,r_{ij}) \in \mathcal{R}_q} \log P_\theta(r_{ij} \mid q). \tag{7}$$

4.4 Answer Selection with Event Relation Graph

The predicted relations form an event relation graph G_q. For multiple choice or yes/no tasks, the candidate set $\mathcal{A}$ is given by the benchmark; for open ended tasks, we sample candidates from the backbone model and add the gold answer during training. We pool pairwise relation distributions into

$$\mathbf{g}_q = \mathrm{Pool}_{i,j}(P_\theta(r_{ij} \mid q)), \tag{8}$$

encode each candidate answer as

$$\mathbf{u}_a = \mathrm{Pool}_{t \in \mathrm{span}(a)} \mathrm{LM}([q; a])_t, \tag{9}$$

and project query and answer representations into a shared space:

$$\tilde{\mathbf{u}}_a = W_u \mathbf{u}_a, \quad \tilde{\mathbf{g}}_q = W_g \mathbf{g}_q. \tag{10}$$

The graph aware score and prediction are

$$S(a, G_q, q) = \mathbf{w}_a^\top [\tilde{\mathbf{u}}_a; \tilde{\mathbf{g}}_q; \tilde{\mathbf{u}}_a \odot \tilde{\mathbf{g}}_q], \tag{11}$$

$$a^* = \arg\max_{a \in \mathcal{A}} S(a, G_q, q). \tag{12}$$

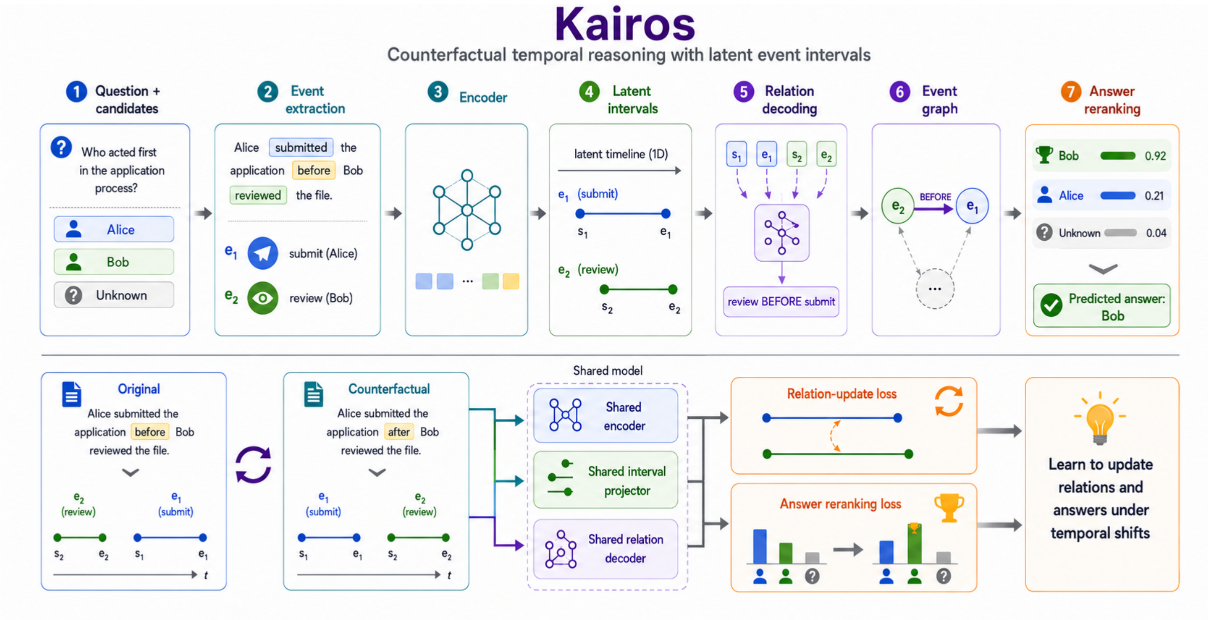

Figure 2: **Overview of Kairos.** The model extracts events from a query, projects event representations into latent intervals, decodes temporal relations into an event graph, and reranks candidate answers. Counterfactual temporal supervision trains the shared encoder, interval projector, and relation decoder to update under event order shifts.

Answer supervision is

$$P_\theta(a \mid q, G_q) = \frac{\exp S(a, G_q, q)}{\sum_{a' \in \mathcal{A}} \exp S(a', G_q, q)}, \tag{13}$$

$$\mathcal{L}_{\text{ans}} = -\log P_\theta(a^* \mid q, G_q). \tag{14}$$

4.5 Counterfactual Temporal Supervision

Given an original query q, we construct a perturbed query q' by changing one target relation while preserving non temporal content. Perturbation templates are split into disjoint train and held-out groups to reduce template leakage. The answer label is updated to a' when the perturbation changes the correct answer. The counterfactual relation loss is

$$\mathcal{L}_{\text{cf}} = -\sum_{(i,j,r'_{ij}) \in \mathcal{R}_{q'}} \log P_\theta(r'_{ij} \mid q'). \tag{15}$$

Unlike representation invariance, this objective trains the model to update relation predictions when event order changes. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{ans}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}} + \lambda_{\text{cf}} \mathcal{L}_{\text{cf}}. \tag{16}$$

Counterfactual construction is summarized in Algorithm 1 in the appendix.

5 Experiments

We evaluate Kairos on temporal subsets from GSM8K, MuSiQue Ans, StrategyQA, and 2Wiki-MultihopQA. We compare direct prompting, chain of thought prompting, self consistency, and a CoT + verifier baseline on Qwen2.5-7B-Instruct unless otherwise stated.

5.1 Experimental Setup

We evaluate on temporal subsets selected from GSM8K, MuSiQue Ans, StrategyQA, and 2Wiki-MultihopQA. Examples are retained only when a temporal dependency is needed for the answer or when a counterfactual order shift has a reliable updated relation.

Table 2 reports subset construction and marker coverage. More than half of retained examples lack reliable relation labels and are used only for answer supervision.

5.2 Relation Update and Stability

We report Relation Update Accuracy after order-changing perturbations and Relation Stability under relation-preserving rewrites. Table 3 shows that Kairos updates target relations more reliably than CoT-based relation extraction.

Table 3 supports the central claim that Kairos improves relation-level temporal structure, not only final answer selection.

5.3 Per-Benchmark Answer Accuracy

Table 1 shows consistent gains over all baselines. Kairos degrades by only 0.6 absolute points under counterfactual shifts on average, compared with 1.7

Table 1: Per-benchmark temporal reasoning results. Standard uses the selected temporal subset. Counterfactual changes one target event order relation while preserving most non-temporal content.

Dataset	Setting	Direct	CoT	Self-Cons.	CoT+Ver.	Kairos
GSM8K	Standard	0.732 ± 0.003	0.828 ± 0.003	0.865 ± 0.002	0.848 ± 0.003	**0.878** ± 0.004
	Counterfactual	0.710 ± 0.003	0.807 ± 0.002	0.844 ± 0.002	0.822 ± 0.003	**0.872** ± 0.004
MuSiQue Ans	Standard	0.254 ± 0.002	0.312 ± 0.002	0.332 ± 0.002	0.331 ± 0.003	**0.358** ± 0.004
	Counterfactual	0.239 ± 0.002	0.298 ± 0.002	0.318 ± 0.002	0.313 ± 0.003	**0.352** ± 0.003
StrategyQA	Standard	0.668 ± 0.003	0.732 ± 0.003	0.752 ± 0.002	0.752 ± 0.003	**0.788** ± 0.005
	Counterfactual	0.646 ± 0.003	0.714 ± 0.002	0.733 ± 0.003	0.730 ± 0.003	**0.782** ± 0.004
2WikiMultihopQA	Standard	0.368 ± 0.002	0.448 ± 0.002	0.468 ± 0.002	0.470 ± 0.003	**0.502** ± 0.004
	Counterfactual	0.359 ± 0.002	0.432 ± 0.002	0.453 ± 0.002	0.455 ± 0.003	**0.496** ± 0.004
Overall	Standard	0.485 ± 0.001	0.559 ± 0.002	0.582 ± 0.001	0.579 ± 0.002	**0.610** ± 0.004
	Counterfactual	0.469 ± 0.002	0.542 ± 0.001	0.565 ± 0.002	0.559 ± 0.002	**0.604** ± 0.003
	Average	0.477	0.551	0.574	0.569	**0.607**

Table 2: Full benchmark sanity checks and temporal subset construction. Full-set numbers use the original dev/test format and Qwen2.5-7B-Instruct. Temporal subset size is the number of examples retained before counterfactual augmentation.

Dataset	Full Direct	Full CoT	Full Self-Cons.	Temporal subset	Marker pair cov.	Relation label cov.
GSM8K	0.732	0.828	0.865	312	68.3%	52.4%
MuSiQue Ans	0.254	0.312	0.332	418	56.0%	39.5%
StrategyQA	0.668	0.732	0.752	365	71.5%	58.1%
2WikiMultihopQA	0.368	0.448	0.468	402	53.7%	41.2%
Average	0.485	0.559	0.582	374	61.8%	47.3%

Table 3: Temporal structure evaluation on the counterfactual partition. CoT relations are obtained either by rule extraction from rationales or by asking the model to output an explicit relation.

Method	Update Acc.↑	Stability↑
CoT rationale + rules	0.328	0.641
CoT relation prompt	0.412	0.703
Kairos	**0.681**	**0.846**

Table 4: Generalization beyond training perturbation templates.

Evaluation split	CoT CF Acc.	Kairos CF Acc.	Update Acc.
In-domain test templates	0.542	**0.604**	0.681
Held-out templates	0.535	**0.597**	0.647
TORQUE/TimeQA-style	0.538	**0.599**	0.632

for CoT, and remains above CoT+Verifier despite using the same reranking setup.

Held-out Templates and Backbone Robustness. Tables 4 and 5 show that the CoT–Kairos gap persists on held-out templates and a second backbone.

5.4 Ablation Confirms the Role of Temporal Structure

We ablate relation supervision, interval projection, counterfactual supervision, and graph-aware answer scoring. Table 6 shows that removing counterfactual supervision or interval endpoints hurts most.

Table 5: Backbone robustness on the overall temporal subset average (Table 1, Average row).

Backbone	Direct	CoT	Kairos
Qwen2.5-7B-Instruct	0.477	0.551	**0.607**
Llama-3.1-8B-Instruct	0.453	0.528	**0.568**

5.5 Error Analysis

Remaining errors mainly come from event extraction mistakes, implicit temporal relations, and noisy counterfactual answer updates.

6 Discussion

The results support explicit event intervals for counterfactual temporal reasoning: gains are largest when event order changes, and relation-level supervision improves update accuracy beyond reranking alone. Remaining limitations include conservative marker-based labels, incomplete event extraction, and implicit temporal relations.

7 Conclusion

We presented Kairos, an interval based framework that decodes event relation graphs and trains relation updates under counterfactual order shifts. Ex-

Table 6: Ablation study of the main components in Kairos. Accuracy is the Standard-setting overall average from Table 1.

Variant	Change from full model	Accuracy	Change
Kairos	Full interval relation graph	**0.610** $\pm$ 0.004	—
w/o Relation Supervision	Train only with answer and counterfactual losses	0.537 $\pm$ 0.003	-0.073
Pair MLP Relation Decoder	Replace intervals with $[\mathbf{h}_i; \mathbf{h}_j; \mathbf{h}_i \odot \mathbf{h}_j]$	0.521 $\pm$ 0.003	-0.089
Single-Point Time Projection	Learn one coordinate per event and no duration	0.549 $\pm$ 0.003	-0.061
w/o Counterfactual Supervision	Remove relation update loss on q'	0.516 $\pm$ 0.003	-0.094
w/o Answer Graph Scoring	Predict answers without graph-aware reranking	0.576 $\pm$ 0.004	-0.034

periments show improved robustness over prompting baselines, especially under counterfactual evaluation.

Limitations

The evaluation is limited to textual temporal subsets and rule-based event extraction. Relation supervision is noisy when temporal relations are implicit, and perturbation diversity remains incomplete.

References

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, and 1 others. 2023. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*.

James F. Allen. 1983. Maintaining knowledge about temporal intervals. *Communications of the ACM*, 26(11):832–843.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, and 1 others. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901.

Wenhu Chen, Xinyi Wang, and William Yang Wang. 2021. TimeQA: A dataset for temporal question answering. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*.

Zheng Chu, Jingchang Chen, Qianglong Chen, Weijiang Yu, Haotian Wang, Ming Liu, and Bing Qin. 2024. TimeBench: A comprehensive evaluation of temporal reasoning abilities in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1204–1228, Bangkok, Thailand. Association for Computational Linguistics.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

Matt Gardner, Yoav Artzi, Victoria Basmov, Jonathan Berant, Ben Bogin, Sihao Chen, Pradeep Dasigi, Dheeru Dua, Yanai Elazar, Ananth Gottumukkala, and 1 others. 2020. Evaluating models' local decision boundaries via contrast sets. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1307–1323.

Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. 2021. Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361.

Rujun Han, Xiang Ren, and Nanyun Peng. 2021. ECONET: Effective continual pretraining of language models for event temporal reasoning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5367–5380, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625.

Divyansh Kaushik, Eduard Hovy, and Zachary C. Lipton. 2020. Learning the difference that makes a difference with counterfactually-augmented data. In *International Conference on Learning Representations*.

Daniel Keysers, Nathanael Scharli, Nathan Scales, Hylke Buisman, Daniel Furrer, Serge Kashubin, Nikola Momchev, Danila Sinopalnikov, Lukasz Stafiniak, Tibor Tihon, and 1 others. 2020. Measuring compositional generalization: A comprehensive method on realistic data. In *International Conference on Learning Representations*.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. In *Advances in Neural Information Processing Systems*, volume 35, pages 22199–22213.

R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. 2019. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3428–3448.

Sewon Min, Eric Wallace, Sameer Singh, Matt Gardner, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2019. Compositional questions do not necessitate multi-hop reasoning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4249–4257.

Qiang Ning, Sanjay Subramanian, and Dan Roth. 2019. An improved neural baseline for temporal relation extraction. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 6203–6209.

Qiang Ning, Hao Wu, Rujun Han, Nanyun Peng, Matt Gardner, and Dan Roth. 2020. TORQUE: A reading comprehension dataset of temporal ordering questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 1158–1172.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, and 1 others. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744.

James Pustejovsky, Jose Castano, Robert Ingria, Roser Sauri, Robert Gaizauskas, Andrea Setzer, and Graham Katz. 2003. TimeML: Robust specification of event and temporal expressions in text. In *Proceedings of the AAAI Spring Symposium on New Directions in Question Answering*.

Yifu Qiu, Zheng Zhao, Yftah Ziser, Anna Korhonen, Edoardo M. Ponti, and Shay B. Cohen. 2024. Are large language models temporally grounded? In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7064–7083, Mexico City, Mexico. Association for Computational Linguistics.

Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912.

Koustuv Sinha, Shagun Sodhani, Jin Dong, Joelle Pineau, and William L. Hamilton. 2019. CLUTRR: A diagnostic benchmark for inductive reasoning from text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 4506–4515.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, and 1 others. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. MuSiQue: Multi-hop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Naushad UzZaman, Hector Llorens, Leon Derczynski, James Allen, Marc Verhagen, and James Pustejovsky. 2013. SemEval-2013 task 1: TempEval-3: Evaluating time expressions, events, and temporal relations. In *Proceedings of the Second Joint Conference on Lexical and Computational Semantics*, pages 1–9.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837.

Siheng Xiong, Ali Payani, Ramana Kompella, and Faramarz Fekri. 2024. Large language models can learn temporal reasoning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10452–10470, Bangkok, Thailand. Association for Computational Linguistics.

Algorithm 1: Counterfactual Temporal Supervision

Data : Training set $\mathcal{D}$, event extractor $\mathcal{E}$, temporal templates $\mathcal{T}$
Result : Augmented training set $\mathcal{D}_{cf}$
foreach *example* $(q, a) \in \mathcal{D}$ **do**
  $E \leftarrow \mathcal{E}(q)$;
  foreach *event pair* $(e_i, e_j) \in E$ **do**
    $r_{ij} \leftarrow \mathrm{DetectRelation}(q, e_i, e_j)$;
    if r_{ij} *is reliable* **then**
      $q' \leftarrow \mathrm{Perturb}(q, e_i, e_j, \mathcal{T})$;
      $r'_{ij} \leftarrow \mathrm{UpdateRelation}(r_{ij})$;
      $a' \leftarrow \mathrm{UpdateAnswer}(a, q, q')$;
      if a' *is reliable* **then**
        Add (q', a', r'_{ij}) to $\mathcal{D}_{cf}$;
return $\mathcal{D}_{cf}$

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, and 1 others. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.

Yucheng Zhou, Xiubo Geng, Tao Shen, Guodong Long, and Daxin Jiang. 2022. EventBERT: A pre-trained model for event correlation reasoning. In *Proceedings of the ACM Web Conference 2022*, pages 850–859. Association for Computing Machinery.

A Counterfactual Construction Details

The rule based detector uses a closed marker list and local syntactic heuristics. Directional markers include before, after, earlier, later, previously, subsequently, followed by, prior to, and once. Overlap and containment markers include during, while, when, as, until, throughout, and meanwhile. We retain a relation only when the two event mentions can be linked to the marker without crossing another explicit temporal connective.

Perturbation templates are split by operation type and surface form. Train templates include direct marker reversal, local event order swaps, and insertion of simple before/after clauses. Held-out templates include paraphrased connectives, subordinate clause movement, and longer clausal reordering. The split contains 18 training template forms and 16 held-out template forms; external TORQUE/TimeQA-style perturbations are kept separate from both groups. UpdateAnswer is applied only when the answer derivation can be deterministically rewritten; otherwise the perturbed example is used for relation evaluation only or discarded from answer training.

DetectRelation maps markers to one of precedes, follows, overlaps, contains, during, or unknown. Perturb changes only one target relation at a time and blocks edits that would introduce two incompatible temporal markers in the same local clause. UpdateRelation applies the inverse relation for directional edits and maps overlap or containment edits to the closest non-overlap relation when the perturbation explicitly moves one event outside the other interval. UpdateAnswer is conservative: numeric answers are recomputed only for templated state-change questions, extractive answers are rewritten only when the answer span is one of the two edited events, and yes/no labels are flipped only for overlap or containment questions with an unambiguous new relation.

B Routing Efficiency Study

Routing is not used as a main accuracy result. We include it only to show the cost tradeoff when structured inference is skipped on examples that appear easy under prompting.

Table 7: Representative counterfactual examples. Queries are shortened for space; the full version stores original query, perturbed query, original relation, updated relation, original answer, and updated answer.

Original query	Perturbed query	Original relation	New relation	Original answer	New answer
Mia saved money before buying a bike. Which happened first?	Mia saved money after buying a bike. Which happened first?	save precedes buy	buy precedes save	saved money	bought bike
The team submitted the draft after revising the figures. What was done earlier?	The team submitted the draft before revising the figures. What was done earlier?	revise precedes submit	submit precedes revise	revised figures	submitted draft
John had 12 apples, gave some away, then bought 5 more. Which operation happened last?	John had 12 apples, bought 5 more, then gave some away. Which operation happened last?	give precedes buy	buy precedes give	bought 5	gave away
The treaty was signed while negotiations continued. Did the events overlap?	The treaty was signed after negotiations ended. Did the events overlap?	sign overlaps negotiate	negotiate precedes sign	yes	no
The museum opened after the renovation. What enabled the opening?	The museum opened before the renovation. What enabled the opening?	renovate precedes open	open precedes renovate	renovation	not renovation
Sarah called the clinic before receiving the test result. Which event came later?	Sarah called the clinic after receiving the test result. Which event came later?	call precedes receive	receive precedes call	received result	called clinic
The package arrived during the storm. Was arrival contained in the storm interval?	The package arrived after the storm. Was arrival contained in the storm interval?	arrive during storm	storm precedes arrive	yes	no
The actor rehearsed, then filmed the scene. Which happened first?	The actor filmed the scene, then rehearsed. Which happened first?	rehearse precedes film	film precedes rehearse	rehearsed	filmed
Nora watered the plants before closing the greenhouse. Which action came later?	Nora watered the plants after closing the greenhouse. Which action came later?	water precedes close	close precedes water	closed greenhouse	watered plants
The software update finished while the backup was running. Did the intervals overlap?	The software update finished after the backup stopped. Did the intervals overlap?	update overlaps backup	backup precedes update	yes	no
The witness arrived after the meeting started. Which event happened first?	The witness arrived before the meeting started. Which event happened first?	start precedes arrive	arrive precedes start	meeting started	witness arrived
Lina mailed the form, then paid the fee. Which step was last?	Lina paid the fee, then mailed the form. Which step was last?	mail precedes pay	pay precedes mail	paid fee	mailed form
The archive opened after the catalog was digitized. What happened earlier?	The archive opened before the catalog was digitized. What happened earlier?	digitize precedes open	open precedes digitize	catalog digitized	archive opened
Priya mixed the batter before heating the oven. Which task was later?	Priya mixed the batter after heating the oven. Which task was later?	mix precedes heat	heat precedes mix	heated oven	mixed batter
The lecture continued while the projector was repaired. Did the events overlap?	The lecture continued after the projector was repaired. Did the events overlap?	lecture overlaps repair	repair precedes lecture	yes	no
Omar submitted the application once the references arrived. What enabled submission?	Omar submitted the application before the references arrived. What enabled submission?	references precede submit	submit precedes references	references	not references
The train left after the doors closed. Which event was first?	The train left before the doors closed. Which event was first?	close precedes leave	leave precedes close	doors closed	train left
The patient rested during the infusion. Was resting contained in the infusion?	The patient rested after the infusion. Was resting contained in the infusion?	rest during infusion	infusion precedes rest	yes	no
The editor approved the article, then scheduled publication. Which action came second?	The editor scheduled publication, then approved the article. Which action came second?	approve precedes schedule	schedule precedes approve	scheduled publication	approved article
Kai downloaded the data prior to training the model. Which step came first?	Kai downloaded the data after training the model. Which step came first?	download precedes train	train precedes download	downloaded data	trained model

Table 8: Selective routing as an efficiency-only analysis. Direct fraction is the proportion of examples sent to direct prompting. Relative cost estimates combine generated tokens and reranking calls, normalized by full Kairos.

Strategy	Direct frac.	Avg. acc.	Rel. cost
Always direct	1.00	0.477	0.31
Confidence gate	0.54	0.537	0.67
Disagreement gate	0.50	0.542	0.71
Random routing	0.50	0.542	0.70
Full Kairos	0.00	**0.607**	1.00